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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 02/17/2005 |
| **PI Name:** | Smith, C. Wayne M.D. | | |
| **Project Title:** | EFFECTS OF ANTI-ORTHOSTATIC SUSPENSION ON INNATE IMMUNITY | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Physiology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77030-2600 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 99-HEDS-03 |
| **Start Date:** | 09/01/2000 | **End Date:** | 01/01/2005 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Li, Zhijie ( Baylor College of Medicine )<br>Rivera, Chantal ( Baylor College of Medicine )<br>Kanwar, Samina ( Baylor College of Medicine ) | | |
| **Grant/Contract No.:** | NAG9-1253 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | A specific aim of this grant was to assess the effects of hind limb unloading (HU) on wound healing. Our previous studies have shown that HU induces liver injury as a result of increased portal endotoxemia, and the current work extends these studies to the evaluation of the inflammatory response associated with wound healing. We have chosen as a model superficial abrasion of the corneal epithelium. This model allows accurate documentation of re-epithelialization and the attendant inflammatory response, including neutrophil, dendritic cell, and Langerhan cell mobilization. Our results reveal a markedly delayed healing and a markedly reduced inflammatory response. | | |
| **Rationale for HRP Directed Research:** | | | |

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| **Research Impact/Earth Benefits:** | The overall objective of this work is to determine a model of altered inflammation and wound healing, reflecting the possible conditions experienced by individuals in zero gravity. The significance of such a model would be to allow definition of possible molecular and cellular mechanisms, and to serve as a basis for development of therapies to promote effective inflammation and wound healing. |
| **Task Progress:** | Introduction: From the critical path roadmap (CPR) the objectives of this grant concerned the following: Do factors associated with flight (stress, environment, microgravity, nutritional status, radiation) affect humoral or cell-mediated immune function, non-specific immunity, mucosal immunity, or immune surveillance capabilities of crewmembers in a manner that exposes them to unacceptable medical risk (disease, allergy, delayed wound healing)? Does the change in fluid distribution away from the extremities affect microcirculation in peripheral tissues, and does this represent a risk to the crew (wound healing, lymphocyte migration, immune surveillance)? How do alterations in neural, endocrine, and metabolic function associated with spaceflight affect cardiovascular functioning and the functioning of other organ systems? (Risk No. 22, Bioastronautics Critical Question Number 7.03; and Risk No. 22, Bioastronautics Critical Question Number 7.10; Risk No. 49, Bioastronautics Critical Question Number 12.02 ).<br>The objectives of this grant dealt specifically with wound healing in the skin and cornea, and the idea that stress and microgravity [as implemented in the ground based model of hindlimb unloading (HU) in rodents] would adversely affect wound healing. As will be pointed out below, we were able to show convincingly that wound healing is delayed in the HU model, and our data indicate important changes in liver function in this model that are consistent with the hypothesis that systemic effects are critical to the functioning of the innate immune system.<br><br>Results of the studies in this project: There are several important finding in these studies that raise potentially critical questions regarding negative effects of space flight on wound healing. They involve wound healing in the skin and wound healing in the cornea.<br><br>Wound healing in the skin: A commonly used experimental model to study wound healing is to place surgical sponges under the skin for a period of time and then determine the extent to which the tissue attempts to heal the wound. There are several ways to measure this response and we chose to determine the amount of collagen produced since collagen is the most important component of a successfully healed wound in adults. We found that there was a greater than 50% reduction in the amount of collagen in 7 and 10 day old wounds from animals exposed to hindlimb unloading (a model developed by NASA scientists that simulates important physiologic aspects the stresses imposed by zero gravity conditions of space flight). These results are consistent with published studies of rats flown in a shuttle flight.<br><br>Wound healing in the cornea: A common concern among astronauts is foreign body injury to the surface of the eye. Normal healing of these types of wounds is critical to the maintenance of excellent vision. Since we found that healing in the skin was reduced, we were concerned that healing in the cornea may also me impaired. Mice subjected to hind limb unloading for 3 weeks were given a superficial injury to the cornea. This involved scraping a small portion of epithelium off of the corneal surface, a common type of superficial injury. These wounds normally heal within 18-24 hours, but there was a significant delay in the healing in the animals exposed to hind limb unloading. Healing time needed to re-grow the epithelium so that the wound was covered was extended by 40 – 50%. What was more troubling, though, was the fact that complete healing (determined by the thickness of the epithelium) was not complete even after 11 days. Normally full thickness is complete within 4 days. Additional concern was raised by another observation. That was that the inflammatory white blood cells that normally come into the wound to protect it from infection were also significantly delayed and reduced in number in the animal exposed to hind limb unloading. From these observations, we are very concerned that if the hind limb unloading model in animals reflects the conditions in space flight, astronauts will be at risk for abnormal healing of superficial surface injuries to the eyes.<br><br>Possible mechanism of reduced wound healing in the skin and eyes: We have found that hind limb unloading causes an increase in bacterial toxin (“endotoxin”) to appear in the blood flowing into the liver. The liver responds to this toxin in several ways. 1) The liver clears the blood of the endotoxin. We could not find endotoxin in the blood that was coming out of the liver. 2) The liver is activated by the endotoxin. We found that the livers began to synthesize new proteins that are released into the blood that is coming out of the liver and distributed throughout the body. Some of these proteins have the ability to alter the inflammatory and healing responses. It is our concern that these proteins may play important roles in the reduced healing in the skin and eyes described above. 3) The liver is injured by the endotoxin. We found inflammatory cells in the livers and measures of cell dealth. The importance of endotoxin is shown by the fact that a strain of mice that are genetically unable to respond to endotoxin did not synthesize new proteins or show evidence of liver injury. Summary of important findings: Hind limb unloading, a NASA model simulating important physiologic stressors of space flight resulted in significant reductions in wound healing in the skin and the cornea. In addition, there was significant reduction in the influx of white blood cells that normally protect the wound from infection. In addition, there were increases in endotoxin in the blood flowing from the intestines into the liver with evidence of liver injury and release of proteins that may have the ability to reduce wound healing. |
| **Bibliography Type:** | Description: (Last Updated: 01/25/2021) |
| **Articles in Peer-reviewed Journals** | Li Z, Rivera CA, Burns AR, Smith CW. "Hindlimb unloading depresses corneal epithelial wound healing in mice." J Appl Physiol. 2004 Aug;97(2):641-7. Epub 2004 Apr 2. PMID: 15064298 , Aug-2004 |
| **Articles in Peer-reviewed Journals** | Rivera CA, Tcharmtchi MH, Mendoza L, Smith CW. "Endotoxemia and hepatic injury in a rodent model of hindlimb unloading." J Appl Physiol. 2003 Oct;95(4):1656-63. PMID: 12794033 , Oct-2003 |
| **Articles in Peer-reviewed Journals** | Rivera CA, Abrams SH, Tcharmtchi MH, Allman M, Ziba TT, Finegold MJ, Smith CW. "Feeding a corn oil/sucrose-enriched diet enhances steatohepatitis in sedentary rats." Am J Physiol Gastrointest Liver Physiol. 2006 Feb;290(2):G386-93. Epub 2005 Oct 13. PMID: 16223947 , Feb-2006 |
| **Presentation** | Smith, C.W.; Burns, A.R.; Li, Z. "CD18 Integrins and Corneal Wound Healing" 12th International Congress of Immunology and 4th Annual conference of FOCIS, Montreal, Canada<br>Jul-2004 |

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| **Presentation** | Li, Z.; Smith, C.W. "Biphasic corneal inflammatory response during corneal epithelial wound healing int he mouse." Association for Research in Vision and Opthalmology (ARVO) Meeting, Ft. Lauderdale, FL.<br>Apr-2004 |
| **Presentation** | Smith, C.W.; Li, Z. "Role of CD11a and CD11b in corneal wound inflammatory process" Association for Research in Vision and Opthalmology (ARVO) Meeting, Ft. Lauderdale, FL.<br>Apr-2004 |
| **Presentation** | Li, Z.; Smith, C.W. "Effects of hindlimb unloading on mouse corneal epithelial wound healing." Experimental Biology 2004, Washington, D.C.<br>Apr-2004 |